

$$U_B = U_{I1} = 4V, U_C = U_{I2} = 1V, i_A = \frac{U_B - U_C}{R}$$

$$U_A = U_B + R \cdot i_A = 7V,$$

$$U_D = U_C - R \cdot i_A = -2V.$$

对 A_3, A_4 ,

$$\begin{cases} \frac{U_A - U_0}{R} = \frac{U_0 - U_3}{R} & U_{03} = U_3 - = U_{3+} = 0V. \\ \frac{U_0 - U_4}{R} = \frac{U_4 - U_0}{R} \\ U_{4+} = U_D \end{cases}$$

$$\therefore U_0 = 2U_D = -4V.$$

(2) 当开关断开时,

$$\text{对 } A_3, \quad U_{03} = -\frac{1}{RC} \int U_A dt = -\frac{20}{0.05} \int U_A dt.$$

对 A_4 , A_4 实现减法运算.

$$\begin{aligned} U_0 &= -\frac{R}{R} + 2U_D - U_{03} = -4 + \frac{20}{0.05} \int U_A dt \\ &= -4 + \frac{140}{0.05} t + U_0(0) = 0.28t \end{aligned}$$

$\therefore$ 当 $U_0 = 0$ 时 $t \approx 28.6\text{ms}$.

6.18 (a) 对 A , $U_0' = -R_3 \left(\frac{U_{I1}}{R_1} + \frac{U_{I2}}{R_2} \right) = 0.1 U_{I3} U_0.$

$$\therefore U_0 = \frac{-10R_3}{U_{I3}} \left(\frac{U_{I1}}{R_1} + \frac{U_{I2}}{R_2} \right).$$

(b) A 实现了反相求和运算, 三个输入电压分别为 $U_I, kU_I^2, k^2U_I^3$

$$\therefore U_0 = -\frac{R_4}{R_1} U_I - \frac{R_4}{R_2} k U_I^2 - \frac{R_4}{R_3} k^2 U_I^3$$